

# Electron configurations

Student: ..... Class: .....

Complete the following tables.

| Name     | Symbol | Atomic number | Electron configuration |
|----------|--------|---------------|------------------------|
| Hydrogen | H      | 1             | 1                      |
| Helium   | He     | 2             |                        |
| Lithium  |        | 3             | 2, 1                   |
|          | Be     |               |                        |
| Boron    |        |               |                        |
|          |        | 6             |                        |
|          |        | 7             |                        |
|          | O      |               |                        |
|          |        |               | 2, 7                   |
| Neon     |        | 10            |                        |
| Sodium   |        |               |                        |
|          | Mg     |               |                        |
|          |        |               | 2, 8, 3                |
|          |        |               | 2, 8, 4                |
|          |        |               |                        |
|          |        |               |                        |
|          | Ar     |               |                        |
|          |        |               |                        |
| Calcium  |        |               | 2, 8, 8, 2             |

| Ion            | Symbol           | Atomic number | Number of electrons | Electron configuration | Matches configuration of |
|----------------|------------------|---------------|---------------------|------------------------|--------------------------|
| <b>Cations</b> |                  |               |                     |                        |                          |
| Lithium        | Li <sup>+</sup>  | 3             | 2                   | 2                      | Helium                   |
| Beryllium      |                  |               |                     |                        |                          |
| Boron          | B <sup>3+</sup>  | 5             |                     |                        |                          |
| Sodium         |                  | 11            | 10                  | 2, 8                   |                          |
| Magnesium      |                  |               |                     |                        |                          |
| Aluminium      | Al <sup>3+</sup> | 13            | 10                  |                        | Neon                     |
| Potassium      |                  |               |                     |                        |                          |
| Calcium        |                  |               | 18                  | 2, 8, 8                |                          |
| <b>Anions</b>  |                  |               |                     |                        |                          |
| Nitride        |                  |               |                     |                        |                          |
| Oxide          | O <sup>2-</sup>  |               |                     |                        |                          |
| Fluoride       |                  | 9             |                     | 2, 8                   |                          |
| Phosphide      | P <sup>3-</sup>  |               | 18                  |                        |                          |
| Sulfide        |                  |               |                     |                        | Argon                    |
| Chloride       |                  | 17            |                     |                        |                          |